Chapter 23

Proficiency Testing, Interlaboratory Comparisons, and Measurement Assurance Programs

PROFICIENCY TESTING

The term *proficiency testing* is often misunderstood when laboratories seeking accreditation hear about it the first time. Many think that its purpose is to assess employees' skills. Although indirectly it is related to the employees' skills and other factors, directly it is a means of determining the laboratory's competence compared to other laboratories performing the same kind of work.

ISO/IEC 17043:2010 defines *proficiency testing* as "evaluation of participant performance against pre-established criteria by means of interlaboratory comparisons."[1] Additionally, ISO/IEC 17043:2010 defines *interlaboratory comparisons* as "organization, performance and evaluation of measurements or tests on the same or similar items by two or more laboratories in accordance with predetermined conditions."[2]

Laboratory proficiency testing is an essential element of laboratory quality assurance. In addition to laboratory accreditation and the use of validated methods, it is an important requirement of the accreditation process. Customers increasingly demand independent proof of competence from laboratories. It is also a requirement if the laboratory is involved in any regulated industry such as nuclear, food, drug, and pharmaceuticals. Laboratories can determine how well they perform against other laboratories and how their measurements compare with others.

For participating laboratories, proficiency testing is a comparison of their individual performance against industry performance. It identifies areas for improvement in measurement and techniques. It may identify best practices by comparing other laboratories in the same field.

For the customers of laboratory services, proficiency testing:

- Establishes confidence and is a demonstration of accreditation
- Helps the customer decide if the laboratory meets its measurement, calibration, and testing requirements
- Acts as a measure for ensuring that the laboratory will continuously meet its quality requirements

Ensuring the validity of results both within one's own laboratory *and* in comparison with other laboratories is a requirement of ISO/IEC 17025:2017. ISO/IEC 17025:2017, Clause 7.7, "Ensuring the validity of results," subClause 7.7.2 states that "[t]he laboratory shall monitor its performance by comparison with results of other

Process control should cover operator training, controlled procedures, and measuring equipment repeatability and reproducibility studies.

Development of measurement uncertainty budgets and measurement uncertainty analysis of the measurement process is another important consideration when reporting the measurement data for proficiency testing. See Chapter 29.

MEASUREMENT ASSURANCE PROGRAMS

It is one thing to compare a laboratory's proficiency against that of another laboratory. But it is equally important in a laboratory's day-to-day operations to ensure that there is a measure of confidence in the testing and calibration work performed.

Some good elements of a measurement assurance program (MAP) include:

- Scheduled, traceable calibration of the laboratory's standards
- Use of check standards
- Comparison of reference standards against check standards and working standards
- Continuous evaluation of the measurement uncertainties associated with the standards and test and calibration processes
- Implementation of a quality system in accordance with recognized international standards such as ISO/IEC 17025:2017
- Controlled, documented test and calibration procedures established under a quality system
- Trained calibration and test technicians (metrologists)
- A formal internal audit program to track the quality of calibration and tests performed
- Control charting and analyzing the results in a timely, proactive manner

Before implementing a MAP, it is important to understand what a laboratory is trying to accomplish. The objectives of a MAP should be defined. The following should be considered as a minimum:

- Select the process to monitor.
- Select the check standard.
- Establish a value for the check standard through traceable calibration or other means (for reference materials).
- Ensure that the check standard is controlled so that its established value is immune to drift due to external factors.
- Monitor and record the process data at a regular, determined interval.
- Chart the data. Control charting is a powerful technique.